VoiceOps 2026 — ZentroMart Support Agent

Team / participant: Fenil · Technical report (measured results only)

1. System overview

The project implements a customer-support chat agent for the fictional ZentroMart ecommerce store. Its retrieval layer indexes current Markdown knowledge-base sections and the supplied product catalog. Hybrid TF-IDF retrieval combines word unigrams/bigrams with character n-grams to help with spelling and ASR-like variation. Retrieved source sections and chunk text are exposed in the interface and saved by the batch runner.

The conversational layer maintains recent context, asks for an order ID and a verification factor before order lookup, allows one retry after a failed verification, and redacts verification input from the stored conversation history. Human handover uses the provided mock ticket tool; the agent confirms creation only when the tool returns created=True. The superseded returns policy is excluded.

2. Evaluation results

Dataset integrity validator: PASS (52 practice questions, 23 conversations / 96 turns, 21 knowledge-base documents and the expected synthetic data tables). Automated unit tests: 7 passed.

Metric	Measured result	Interpretation
Practice questions indexed	52	Queries use asr_text when available.
Answerable questions	47	From supplied answerable labels.
Gold evidence hit@5	36 / 47 = 76.6%	At least one expected evidence identifier appears in top-five retrieved chunks.
Mean reciprocal rank	0.4989	Calculated over the 47 answerable questions.
Unsupported questions	5	Retrieved chunks may still appear; model abstention needs separate testing.

These are retrieval metrics, not generated-answer accuracy. Step 1 prompt-only accuracy, Step 2 answer accuracy, full conversation scoring and voice quality have not yet been measured. The CSV and JSON retrieval outputs are included in the project's evaluation/ folder.

3. What was improved and remaining limitations

Improvements made and tested: product-catalog indexing; hybrid word/character retrieval; exclusion of superseded policy documents; a bounded order-verification retry; explicit handling of malformed verification input; redaction of verification digits from conversation history; per-audio-file session reset to avoid cross-customer context leakage; and automated tests for these behaviors and ticket handover.

The main observed retrieval limitation is that 11 of the 47 answerable questions do not retrieve a gold evidence identifier in the top five. Unsupported questions can still retrieve superficially similar chunks, so abstention behavior must be checked against the actual generated answers. Product facts are retrieved from the supplied product CSV, while policy claims should be grounded in current KB documents.

4. Reproducibility and voice status

Chat: run `streamlit run app.py`. **Retrieval evaluation:** run `python evaluate.py --top-k 5 --out-dir evaluation`. **Tests:** run `python -m unittest discover -s tests -v`. **Batch voice command:** `python run.py --audio-dir sample_audio --out-dir outputs`. Configure DATASET_DIR and GNANI_API_KEY in a local .env file; secrets are excluded from submission.

Gnani STT/TTS integration is implemented against the SDK calls specified in the capstone brief, but end-to-end audio has not been run in this development environment because no Gnani API key was available. Before final submission, provide your own key locally, test a recorded voice turn, publish the repository, and add the required ≤2-minute demo video. Do not represent these pending checks as completed.